

CONTINUOUS ASSESSMENT 2

Multivariate Analysis

Joshua Orelowitz and Keeran Moodley

Revised on: March 1, 2026

Abstract

An exploration of the singular value decomposition (SVD) and its applications in multivariate data analysis

Table of contents

Question 1 3

Question 2 3

Question 3 3

Question 4 4

Question 1

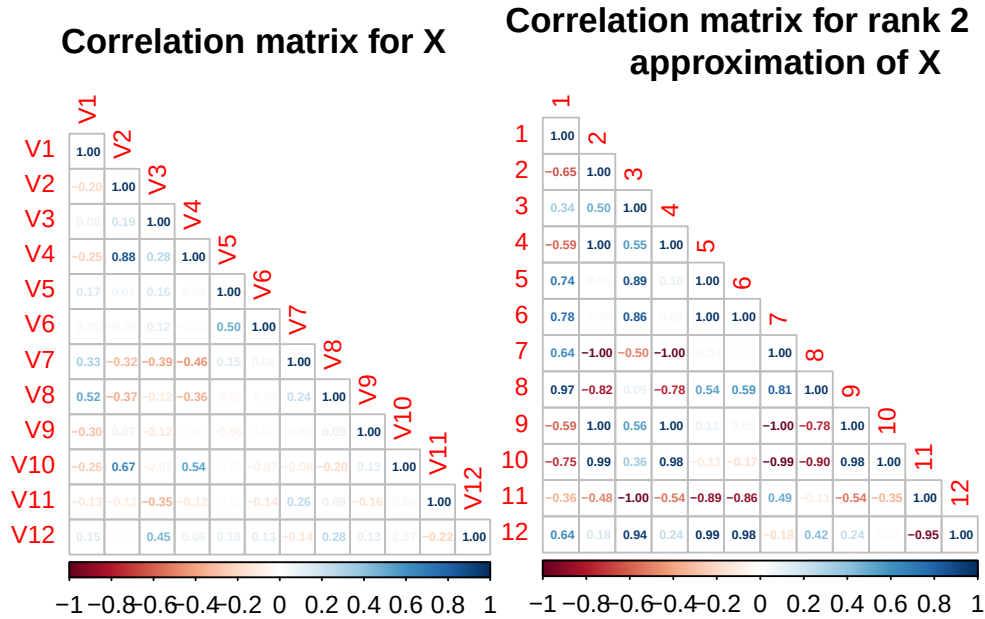
We now compute the lower rank approximations for all ranks. From this we select the case where $\text{rank}(\tilde{X}_4) = 4$ and compute the error.

Table 1: Errors for the rank 4 approximation

1	2	3	4	5	6	7	8	9	10	11	12
0.012	-0.011	0.132	0.015	-0.042	0.012	0.01	-0.021	0.061	-0.043	0.138	-0.038

Question 2

We now compare the correlation structure for the full matrix and that of the rank 2 approximation.

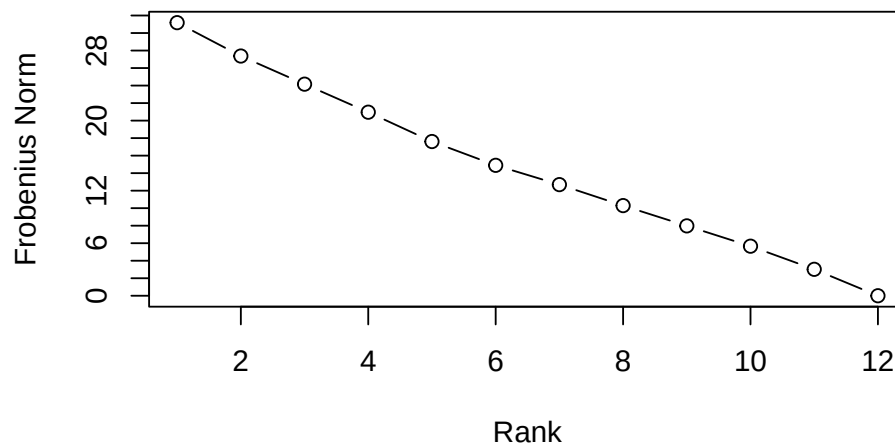


The correlation matrix for X shows the correlation structure between all variables. The correlation matrix for the rank 2 approximation of X only explains the variation found by the first 2 singular components. Our rank 2 approximation starts to pick up on some of the stronger correlations such as the correlation between V4 and V2, V10 and V2. However it also seems to be picking up correlation that as the ranks develop seems to erode, such as showing a correlation of 1 for V9 and V2 which in X only has a correlation of 0.07. It also gives a correlation of -1 between V9 and V7 which in truth seem to actually be very uncorrelated with a value of almost 0. More often than not our rank 2 approximation is picking up correlation that does not seem to truly exist. This can be expected as we are compressing the data into a 2 dimensional space. Therefore I would say that a rank 2 approximation is not accurate and that we need to do a higher rank approximation to accurately capture the correlation structure in X.

Question 3

We now compute the Frobenius norm of the delta matrices, $\Delta_k = X - \tilde{X}_k$, for $k = 1, 2, \dots, 12$ corresponding to the respective ranks.

Frobenius norm for different K values

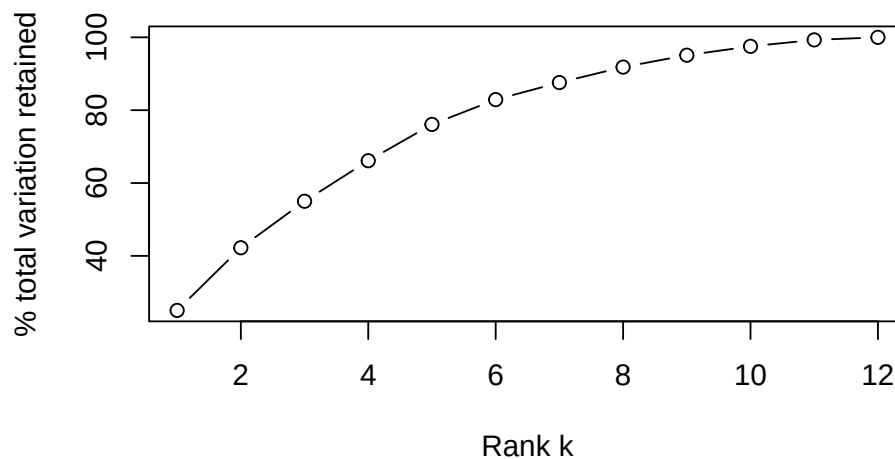


We notice that as the rank of the approximation increases the Frobenius norm of the difference between the original matrix and its approximation decreases. This is to say that as the rank of the approximation increases the approximation becomes closer to the original matrix until they are exactly the same - as the rank increases as does the accuracy of the approximation.

Question 4

We now compute and plot the percentage of variation retained by the lower rank approximations and examine how they evolve as the rank for the approximation varies.

Variation retained in X for different K values



We can see that the percentage of variation retained of the total X increases with the rank of our SVD. At first it is very steep but seems to flatten out at around a rank of 9 which explains around 90% of the variation. This means that most of the variation is captured in the first 9 singular components. Higher rank approximations add relatively low amounts of variation.